

Buildbarn Portal Demo

Buildbarn Meetup 2026

Agenda

- Overview
- Scheduler UI
- Browser UI
- Build Event Stream UI
- Completed Actions UI
- Interactive demo

Interactive Demo

- https://github.com/meroton/buildbarn-meetup-demo

Agenda

- **Overview**
- Scheduler UI
- Browser UI
- Build Event Stream UI
- Completed Actions UI
- Interactive demo

Overview

- A Buildbarn web UI
- Used for visualizing Bazel builds and Buildbarn cluster information
- Replacement for bb-browser and bb-scheduler web UI
- Parses the Build Event Stream and displays logs, metrics etc.

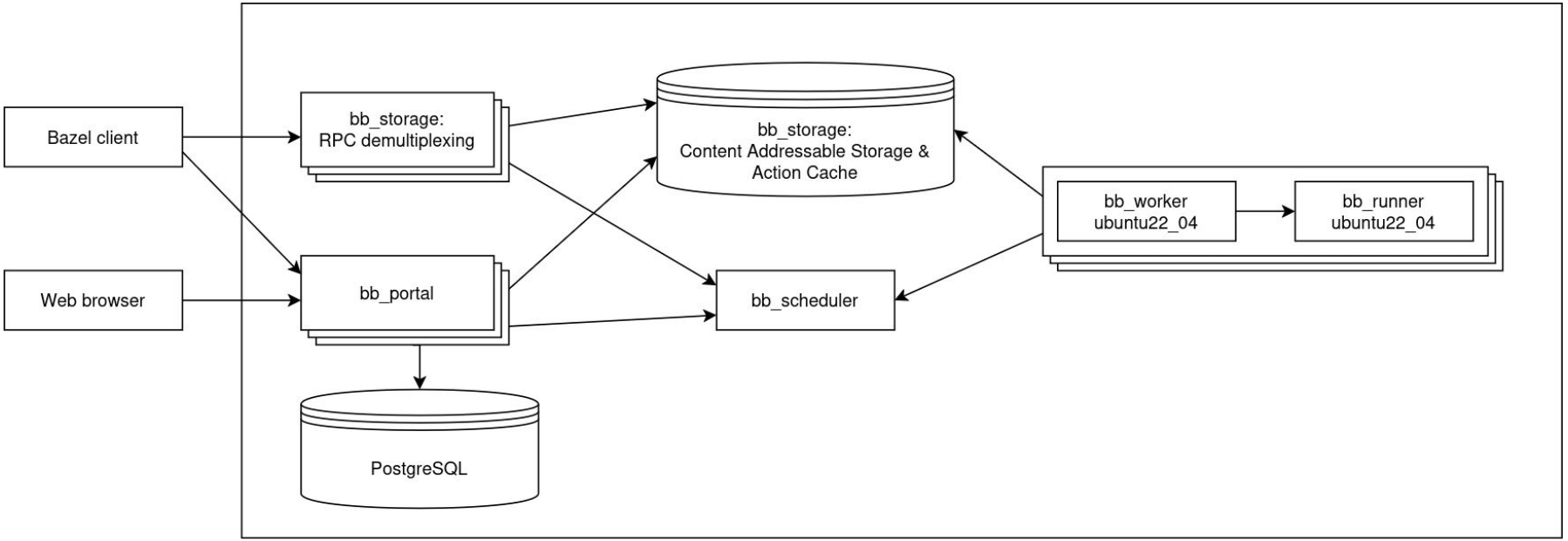
Overview - Tech Stack

- Frontend - React Typescript SPA, Tanstack Router
- Backend - Go, using Buildbarn packages
- PostgreSQL with a GraphQL API
- gRPC for talking to the scheduler and the remote execution cache
- Single container image that serves both frontend and backend

Overview - Deployment

- Scales to thousands of developers, millions of invocations with single digit number of cores
- Stateless - can scale horizontally
- Operationally simple to deploy
  - PostgreSQL database needed for Build Event Stream data
  - Buildbarn cluster needed for Browser and Scheduler UI

Overview - Example Setup

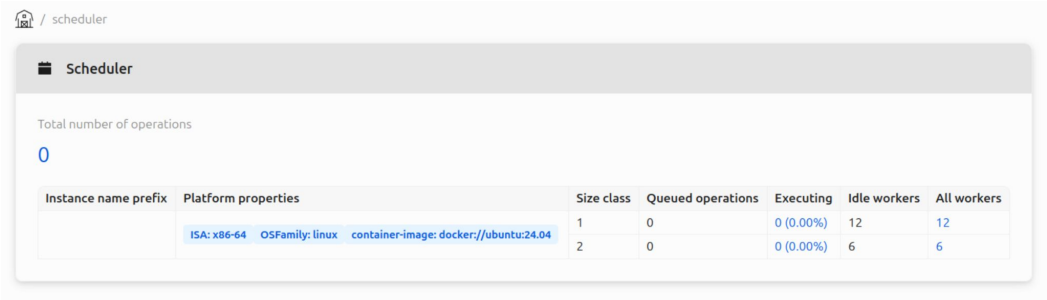


Agenda

- Overview
- **Scheduler UI**
- Browser UI
- Build Event Stream UI
- Completed Actions UI
- Interactive demo

Scheduler UI

- Visualizes the state of remote execution in the cluster
  - Platform queues
  - Workers
  - Ongoing operations
- Uses gRPC, proxied through the backend, to talk to the scheduler BuildQueueState API

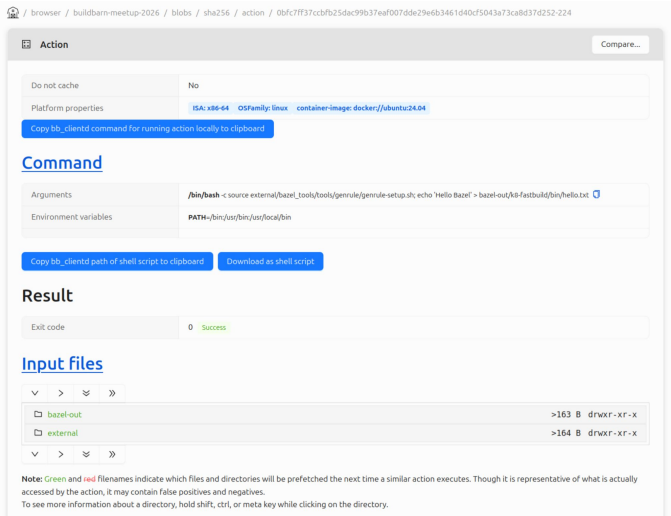


Agenda

- Overview
- Scheduler UI
- **Browser UI**
- Build Event Stream UI
- Completed Actions UI
- Interactive demo

Browser UI

- Visualization of objects from the Buildbarn cache
  - Actions
  - Input and output files
  - Historical execute response
  - etc.
- Uses gRPC, proxied through the backend, to talk to the Buildbarn cache

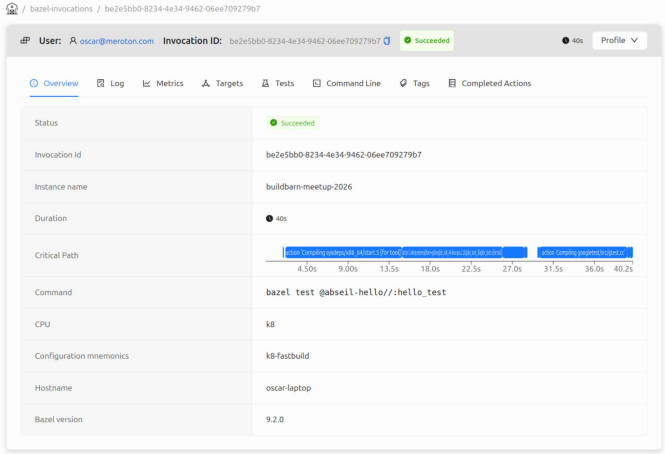


Agenda

- Overview
- Scheduler UI
- Browser UI
- **Build Event Stream UI**
- Completed Actions UI
- Interactive demo

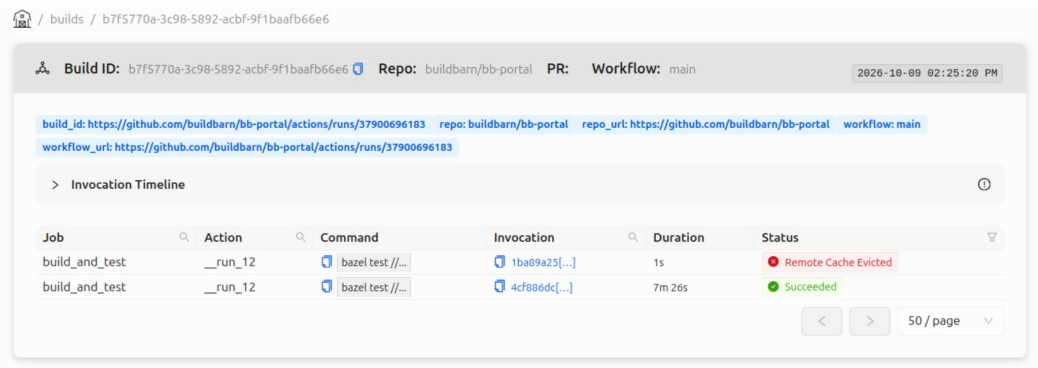
Build Event Stream UI

- Bazel sends the Build Event Stream to bb-portal using the flag `--bes_backend`
- Bazel prints a link to bb-portal for every invocation
- Parsed and stored in PostgreSQL database
- Served to the frontend via a GraphQL API



Build Event Stream UI

- Groups invocations together into Builds
- Can be used to track all invocations from a CI job
- Configurable to suit most CI systems

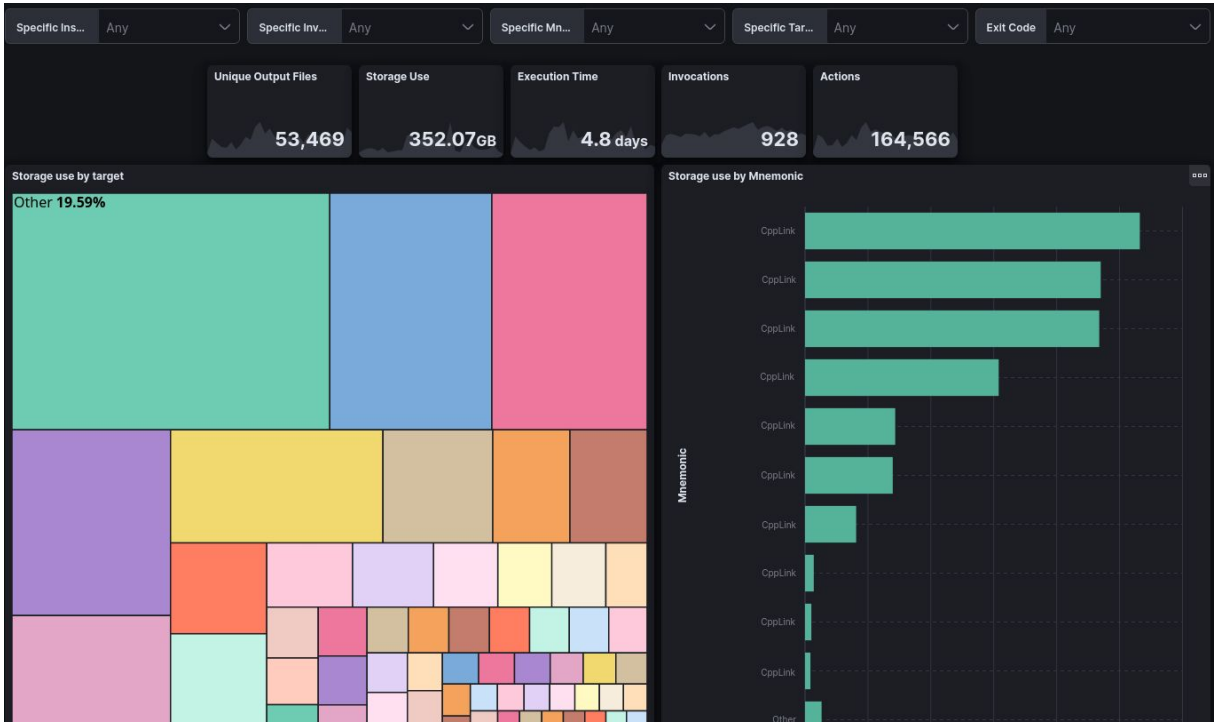


Agenda

- Overview
- Scheduler UI
- Browser UI
- Build Event Stream UI
- **Completed Actions UI**
- Interactive demo

Completed Actions UI

- Completed actions logger
- Coming soon™ to bb-portal



Agenda

- Overview
- Scheduler UI
- Browser UI
- Build Event Stream UI
- Completed Actions UI
- **Interactive demo**

Interactive Demo

- https://buildbar.meroton.com
- https://github.com/meroton/buildbarn-meetup-demo
- Small cluster - Please don’t load test it
- We will remove all personal information after the demo
- Instructions in the README



Questions?

Isak Stenström

isak@meroton.com

Oscar Falk

oscar@meroton.com